

PalmSource™ User Interface Development

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Agenda

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- Advice
- A little Zen of Palm™
- Twelve steps to make your application easy to use by User Interface Development

Advice—Building the User Interface Development Team

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- User interface development comes from a cross functional team
  - Marketing, engineering, product design, experts, users, designers, human factors, usability, documentation, Q&A, Sales, Call Centers, etc.
  - Document all interactions with users and experts
  - Keep the end user in mind at all times
    - Give them what they need not what they want
  - Agree to a plan of action and schedule
  - Take the time to do it right…the first time

Advice—Easy to Use

- To create “easy to use” software/hardware you understand the user and the process of the user’s tasks
  - The user profile
  - The tasks required to get their job done
  - “Just in time information” for the user
  - Understand the limitations on the platform
  - Heavy user involvement
    - Experts
    - Basic user (Lots and lots of them!)
    - Names and numbers of actual people

Advice—Hard to Deliver

- Delivering an “easy to use” product is difficult
  - Everyone wants it now
  - Most are willing to skip steps to get the product out the door sooner
  - Engineering wants to start coding
  - Marketing wants to see the “final version” in the initial stages of UI development
  - Everyone is in such a rush they overlook who it’s for and what they are trying to do
  - Most times user interface development teams get involved too late in the project to make an impact on usability or design of the product

Advice—Pitfalls

- Who is in charge of the project
  - Who are the key players on the project
- What is the real deadline/schedule
  - Are all members signed up to the same agenda/schedule?
  - What's the bare minimum needed meet the schedule?
- Feature Creep
  - Make sure good ideas are written down!
- Before all presentations make sure you recite the goals of your project

Advice—Education

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- Educate yourself and others about user interface development
  - Join user groups, BayChi, or other human computer interaction groups, read books, go to conferences, talk to designers
  - Collect examples of great designs
- Educate your leaders
  - Time and money is needed to build a better UI
  - Put a standard process in place so that it can be successfully repeated (saves time and money)
  - Once the project is underway changing course will affect the timeline and budget

Zen of Palm™
The PC Is Like a Home

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- Contains everything you need
- Plenty of room and power
- More features is always better
- I would rather live in a mansion than a condo

Zen of Palm™
The Handheld Is Like a Car

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- Doesn’t have everything a home has
- Limited of room and power
- More features limit speed and flexibility
- I would rather drive an Acura than a Winnebago

Zen of Palm™ Usage Patterns

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- Handhelds are used for quick bursts of information
  - Check your calendar
  - Look up a phone number
  - Jot down a memo
- PCs are used for longer periods of time
  - Work on a spreadsheet
  - Compose a letter
  - Draw or illustrate an idea

Zen of Palm™
Why a PC Approach Won't Work

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- Desktop application at your fingertips
- Provide robust solutions
  - Carpet bomb approach
- Give Customers what they ask for
- Focus on adding and enhancing features

Zen of Palm™ Handheld Approaches

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- Clever solutions instead of brute force
- Robust applications are cop outs
- Focus on doing a few key things really well
- Power = ability to get the job done
- Simplicity is power

Zen of Palm™
Determine the Need

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- Solve a problem
  - Right—what is the user trying to do?
  - Wrong—How do I enhance this feature?
- Focus on the 80/20 rule
  - 80% of the time customers only use 20% of the features in an application
- Example: Business card beaming
  - What one, two, or three things do your customers use all the time? Make those just one tap

Zen of Palm™
Scale the Problem

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- Define specifications like you pack a back pack
  - Start from 0 and build to 100%
- Ignore your customers
  - Give them what they need not what they want
- Let the PC do the heavy lifting
  - The customer has a PC or a Macintosh
- Example: Expense application

Zen of Palm™ Implementation

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- Fast and easy
  - Minimize clutter
  - Remember “less is more”
  - Minimize steps for the features used frequently
  - Consistency = familiarity = easier
  - UI Guidelines matter
    - Limit the number of speed bumps
  - Intuitive once you figure it out
    - Learned usage is OK

User Interface Development

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- Outlined next are 12 steps to creating an “easy to use” application
- In the beginning of your job you won’t have enough time to complete all the steps; But after you have been able to implement some of these steps, you and your coworkers will see their value
- Over time you will be able to take a project through all 12 steps, that will be the project that lands in your portfolio

User

- The goal is to develop an understanding of the user and tasks
  - User Analysis
  - Task Analysis
  - Develop a set of test scenarios
  - Write usability requirement

Interface

- The goal is to optimize the usefulness of your design
  - Identify Key functions and prioritize
  - Develop conceptual designs

Development

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- The goal is to optimize the usability of your interface
  - Develop alternative interface designs
  - Early evaluation and usability testing
  - Perform usability inspection
  - Perform advanced usability testing
  - Write user interface specifications
  - Field testing

Three Pillars

- There are three pillars to the Zen of User interface development
  - Teaching
  - Practice
  - Enlightenment
    - I have taught you steps to create “easy to use” applications; it is up to all of us to understand it so we can all at some point teach it
    - Practice what you teach
    - Enlightenment doesn’t just come after Teaching and Practice, it arrives at anytime
    - Be ready to embrace it, and document it!


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